

WEEK 4 PROJECT SUMMARY

HealthConnect Appointment Attendance and No-Show Analysis

Dataset: Health Connect Appointment Data

By

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SPECIFIC PROBLEM TO BE ADDRESSED

The Data Analytics track will focus on understanding appointment attendance and no-show patterns at HealthConnect Clinic. The analysis aims to identify factors associated with missed appointments and provide data-driven insights that can support improved appointment management and patient attendance.

RESOURCES USED

The analysis was based on:

- HealthConnect Appointment Dataset
- HealthConnect Data Dictionary
- Power Query for initial data-quality assessment and preparation

KEY OBSERVATIONS FROM WEEK 4

The initial review identified the following key observations:

- The dataset contains 5,000 appointment records and 18 variables covering patient, appointment, booking, previous appointment history, reminder, accessibility, and outcome information.
- Two date fields, `appointment_date` and `booking_date`, initially required conversion from Text to Date.
- Limited missing values were identified for `distance_to_clinic_km` and `waiting_time_minutes`.
- No duplicate records or duplicate appointment IDs were identified.

- No obvious invalid or implausible values were identified during the initial validity checks.
- Seven variables were identified as particularly relevant to investigating appointment attendance and no-shows: appointment_outcome, booking_lead_days, previous_no_shows, appointment_type, reminder_sent, distance_to_clinic_km, and age_group.

PROPOSED APPROACH

The analysis will begin by examining overall appointment outcomes and no-show patterns. It will then investigate associations between no-shows and relevant patient, appointment, booking, reminder, and accessibility factors.

The proposed KPIs, No-Show Rate, Attendance Rate, Repeat No-Show Rate, and Reminder Attendance Rate, will be calculated and evaluated. Findings will then be used to develop business insights and recommendations.

KEY CONSIDERATIONS THAT MAY AFFECT THE PROJECT

The analysis will take into account:

- Limited missing values in some variables.
- The fact that the dataset may not capture every reason for missed appointments.
- The need to interpret relationships between variables as associations rather than proof of causation.
- The importance of accurate "appointment_outcome" data and consistent use of the Data Dictionary.

PROPOSED FOCUS FOR WEEK 5

The focus for Week 5 will be to conduct the proposed analysis of appointment attendance and no-show patterns. This will include:

- Calculating the selected KPIs.
- Comparing relevant patient and appointment groups.
- Identifying meaningful attendance and no-show patterns.
- Developing data-driven insights to support HealthConnect's objective of reducing missed appointments.